Sec: OSR.IIT_*COSC(MODEL-A&APEX) | GTA-2(P2) | Date: 29-03-26
Time: 3 Hrs | 2020_P2 | Max. Marks: 198

KEY SHEET
MATHS

1	**5**	2	**6**	3	**7**	4	**4**	5	**1**
6	**8**	7	**ACD**	8	**ABC**	9	**AC**	10	**ABC**
11	**BD**	12	**ABD**	13	**0**	14	**2**	15	**1344**
16	**33.33**	17	**96**	18	**2**				

PHYSICS

19	**8**	20	**4**	21	**4**	22	**8**	23	**5**
24	**3**	25	**CD**	26	**ACD**	27	**C**	28	**D**
29	**BC**	30	**ABC**	31	**3**	32	**2.40 to 2.60**	33	**13.5**
34	**1.25 to 1.35**	35	**1.01 to 1.08**	36	**0.40 to 0.60**				

CHEMISTRY

37	**2**	38	**2**	39	**6**	40	**5**	41	**6**
42	**5**	43	**AC**	44	**AD**	45	**ABD**	46	**BC**
47	**ACD**	48	**AC**	49	**10**	50	**25**	51	**6**
52	**14**	53	**4.95 to 5.05**	54	**12.24**				

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THE NARAYANA METHOD
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SOLUTIONS
MATHS

1. $(z_1+iz_2)^3 = z_1^3+3z_1^2iz_2+3z_1i^2z_2^2+i^3z_2^3$

$= z_1^3-3z_1z_2^2+i\left(3z_1^2z_2-z_2^3\right)$

$(z_1+iz_2)^3 = 2+11i$ also $(z_1-iz_2)^3 = 2-11i$

$(z_1+iz_2)(z_1-iz_2) = z_1^2+z_2^2$

$(z_1+iz_2)^3(z_1-iz_2)^3 = \left(z_1^2+z_2^2\right)^3$

$(2+11i)(2-11i) = \left(z_1^2+z_2^2\right)^3$

$4+121 = \left(z_1^2+z^2\right)^3$

$z_1^2+z_2^2 = 5$

2. Let P_n is prob that sum of n dice is divisible by 7.

Also q_n is prob……is not divisible by 7.

If $n-1$ dice sum is divisible by 7 then $\{1,2,3,4,5,6\}$ number on n^{th} dice

$\Rightarrow$ total sum is not divisible by 7

$\Rightarrow P_{n-1} = 0$ (when sum of $n-1$ dice is divisible by 7)

If $n-1$ dice sum is not divisible by 7 its probability is q_{n-1}

Sum of $n-1$ not divisible by 7 may have remaining 1, 2, 3, 4, 5, 6

The sum of $n-1$ dice was not divisible by 7 (q_{n-1}). There are 6 possible non-zero remainders. For any specific remainder, there is exactly **one** face on the n-th die that will bring the total sum to a multiple of 7.

Example: If the sum of $n-1$ dice has remainder 1, we need the n-th die to be 6.

Example: If the sum of $n-1$ dice has remainder 2, we need the n-th die to be 5.

Since there are 6 faces on the die, the probability of "fixing" the sum is $1/6$.

$P_n = P_{n-1}.O+q_{n-1}.\frac{1}{6}$

$P_n = \frac{q_{n-1}}{6};\ p_n+q_n = 1$

$p_n = \frac{1-p_{n-1}}{6}$ $\qquad p_{n-1}+q_{n-1} = 1$

$q_{n-1} = 1-p_{n-1}$

$n=0\ \ p_0 = 0$

$n=1\ \ p_1 = 0$

$n=2\ \ p_2 = \frac{1-0}{6} = \frac{1}{6}$

$n=3\quad p_3 = \frac{1-p_2}{6} = \frac{1}{6}-\left(\frac{1}{6}\right)^2$

$n=4\ \ p_4 = \frac{1}{6}-\frac{1}{6}p_3 = \frac{1}{6}-\frac{1}{6}\left(\frac{1}{6}-\left(\frac{1}{6}\right)^2\right) = \frac{1}{6}-\frac{1}{6^2}+\frac{1}{6^3}$

$n=10\ p_{10} = \frac{1}{6}-\frac{1}{6^2}+\frac{1}{6^3}-\frac{1}{6^4}+\frac{1}{6^5}-\frac{1}{6^6}+\frac{1}{6^7}-\frac{1}{6^8}-\frac{1}{6^9}$

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$$=\frac{\frac{1}{6}\left(1-\left(-\frac{1}{6}\right)^9\right)}{1+\frac{1}{6}}=\frac{1}{6}\cdot\frac{6}{7}\left(\frac{6^9+1}{6^9}\right)=\frac{1}{7}\left(\frac{6^9+1}{6^9}\right)$$

$$p=\frac{1}{7}\left(\frac{6^9+1}{6^9}\right)\frac{1}{p}=7\left(\frac{6^9}{6^9+1}\right)$$

$\frac{6^9}{6^9+1}$ is slightly smaller then 1

$$\frac{1}{p}=7(0.9999....)=6.9999....$$

$$\left[\frac{1}{p}\right]=[6.999]=6$$

3. $2b=a+c\Rightarrow a-2b+c=0$

$ax+by+c=0$

$x=1, y=-2$

$\Rightarrow \alpha=1, \beta=-2$

$x^2+y^2-2\alpha x-2\beta y+\alpha^2+\beta^2-\gamma=0$

$2g_1g_2+2f_1f_2=c_1+c_2$

$2(-\alpha)(-2)+2(-\beta)(-2)=(\alpha^2+\beta^2-\gamma)-1$

$4\alpha+4\beta=\alpha^2+\beta^2-\gamma-1\Rightarrow 4-8=1+4-\gamma-1$

$-4=4-\gamma\Rightarrow\gamma=8$

$1-2+8=7$

4. $|A-\lambda I|=0$

$\lambda^3-c_1\lambda^2+c_2\lambda-c_3=0$

$c_1=tr(A),\ c_2=\frac{1}{2}\left((tr(A))^2-tr(A^2)\right)$

$c_3=\det(A)$

$c_1=\text{tr}(A)=3$

$c_2=\frac{1}{2}(3^2-5)=\frac{4}{2}=2$

Using the Cayley-Hamilton Theorem, $A^3-c_1A^2+c_2A-c_3I=O$. Taking the trace:

$$\text{tr}(A^3)-c_1\text{tr}(A^2)+c_2\text{tr}(A)-3c_3=0$$
$$15-3(5)+2(3)-3c_3=0$$
$$15-15+6-3c_3=0$$
$$3c_3=6\implies c_3=2$$

So, $\det(A)=2$.

Step 2: Evaluate the target expression.

The characteristic equation is $A^3-3A^2+2A-2I=O$.

Rearranging the terms:

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$$A(A^2 - 3A + 2I) = 2I$$
$$A^2 - 3A + 2I = 2A^{-1}$$

Taking the determinant of both sides:

$$\begin{aligned}\det(A^2 - 3A + 2I) &= \det(2A^{-1}) \\ &= 2^3 \det(A^{-1}) \\ &= \frac{8}{\det(A)} \\ &= \frac{8}{2} = 4\end{aligned}$$

5.

The function $f(x)$ is given by:
$f(x) = |x| + \left|\frac{x}{3} - 1\right| + \left||x| \cdot \left|\frac{x}{3} - 1\right|\right|$

- The absolute value function $u(x) = |x|$ is continuous for all $x \in \mathbb{R}$.
- The linear function $v(x) = \frac{x}{3} - 1$ is continuous for all $x \in \mathbb{R}$, and its absolute value $|v(x)|$ is also continuous everywhere.
- The third term is the absolute value of the product of two continuous functions, which is itself continuous.
- Since $f(x)$ is the sum of three continuous functions, it is continuous for all $x \in \mathbb{R}$.

Thus, the number of points of discontinuity of $f(x)$ is:
$\alpha = 0$

Determination of β

We are given $g(x) = |2 - |x - 2||$ and $h(x) = \frac{|\sin x|}{\sin x}$ for $x \neq 0$. We need to find the sum of the roots of $g(x) - h(x) = 0$ in the interval $[-2, 2]$.

Analysis of $g(x)$ on $[-2, 2]$:
For $x \in [-2, 2]$, the expression $x - 2$ lies in the range $[-4, 0]$. Therefore, $|x - 2| = -(x - 2) = 2 - x$.
Substituting this into $g(x)$:

$$\begin{aligned}g(x) &= |2 - (2 - x)| \\ &= |x|\end{aligned}$$

Analysis of $h(x)$ on $[-2, 2]$:
The function $h(x) = \frac{|\sin x|}{\sin x}$ is the signum function of $\sin x$.

- For $x \in (0, 2]$, $\sin x > 0$, so $h(x) = \frac{\sin x}{\sin x} = 1$.
- For $x \in [-2, 0)$, $\sin x < 0$, so $h(x) = \frac{-\sin x}{\sin x} = -1$.
- At $x = 0$, $h(x)$ is undefined.

Solving $g(x) = h(x)$:

- **Case 1:** $x \in (0, 2]$
  $|x| = 1 \implies x = 1$ or $x = -1$
  Since $x \in (0, 2]$, the only root is $x = 1$.
- **Case 2:** $x \in [-2, 0)$

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$|x| = -1$

Since the absolute value of a real number is always non-negative, there are no solutions in this interval.

The only root of the equation in the interval $[-2, 2]$ is $x = 1$. Thus, the sum of the roots is:

$\beta = 1$

Calculation of $\alpha + \beta$

Using the values found above:

$$\alpha + \beta = 0 + 1$$
$$= 1$$

6. Let $x^x = t$

$x \to 0^+ \Rightarrow x^x = 1$

$$L = \lim_{t \to 1} \frac{e^{t-1} - t}{\left(t^2 - 1\right)^2}$$

Let $t - 1 = u$

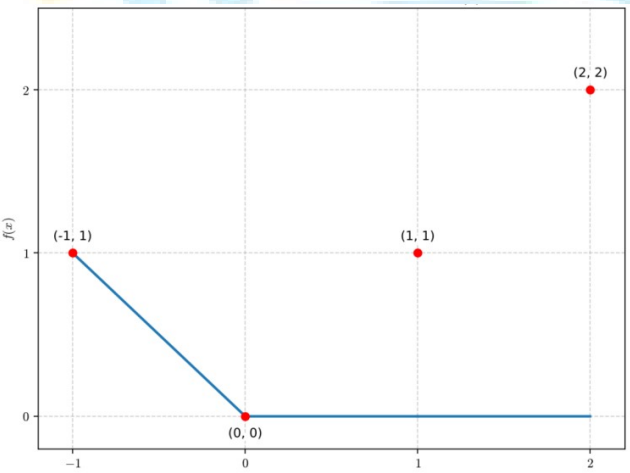
$u \to 0$

$$L = \lim_{u \to 0} \frac{e^4 - u - 1}{(t-1)^2 (t+1)^2}$$

$$= \lim_{4 \to 0} \frac{e^4 - u - 1}{u^2 (u+2)^2} \quad L = \frac{1}{8}$$

$$\frac{1}{L} = 8$$

7.
$$f(x) = \begin{cases} -x & \text{if } -1 \le x < 0 \\ 0 & \text{if } 0 \le x < 1 \\ 1 & \text{if } x = 1 \\ 0 & \text{if } 1 < x < 2 \\ 2 & \text{if } x = 2 \end{cases}$$



8. $ax^2 + 2hxy + by^2 + 2gx + 2fy + c = 0$

$9x^2 + 24xy + 16y^2 - 46x - 28y - 3 = 0$

$h^2 - ab \Rightarrow 144 - 9.16 = 0 \Rightarrow$ parabola

$y^2 = 4ax; y = 0$ i.e x axis

Axis $z = 0$ i.e g axis tangent at vertex

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$(3x+4y)^2 = 46x+28y+3$

$3x+4y=0 \Rightarrow$ represents a line passing through origin for axis of parabola add λ inside square

$(3x+4y+\lambda)^2 = (3x+4y)^2 + 2(3x+4y)+\lambda^2$

$= (46x+28y+3)+(6x+8y)+\lambda^2$

$\underbrace{(3x+4y+\lambda)^2}_{Axis} = \underbrace{x(46+6\lambda)+y(28+8\lambda)+3+\lambda^2}_{\text{tangent at vertex}}$

T_v and axis are perpendicular to each other

$\left(\frac{-3}{4}\right)\left(\frac{-(46+6\lambda)}{28+8\lambda}\right) = -1$

$138+18\lambda = -112-32\lambda$

$250 = -50\lambda$

$\lambda = -5 \ldots\ldots\ldots\ldots in(1)$

$(3x+4y-5)^2 = 16x-12y+28 \Rightarrow (3x+4y-5)^2 = 44x-3y+7$

Vertex is P.O.I $3x+4y-5=0$

$4x-3y+7=0$

$12x+16y-20=0$

$12x-9y+21=0$

$25y = 41$

$y = \frac{41}{25}, \; x = \frac{-13}{25}$

Option B is correct

$\left(\frac{3x+4y-5}{5}\right)^2 25 = 4.5\frac{4x-37+7}{5}$

$\left(\frac{3x+4y-5}{5}\right)^2 = \frac{4}{5}\left(\frac{4x-3y+7}{5}\right)$

$y^2 = \frac{4}{5}x$

$a = \frac{1}{5}$

Directrix $x=-a$

$\frac{4x-3y+7}{5} = -\frac{1}{5}$

$4x-3y+8=0$

$x=-2 \Rightarrow -8-3y+8=0$

$y=0$

C is correct

Focus x = 0 and y = 0 $\Rightarrow \frac{4x-3y+7}{5} = \frac{1}{5}$

$\Rightarrow 4x-3y+6=0$

$4x-3y+6=0$ and $\frac{3x+4y-5}{5}=0$

$4x-3y+6=0$ and $3x+4y-5=0$

Focus $\left(\frac{-9}{25}, \frac{38}{25}\right)$

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9.

$$y=(\cos x)^x$$

$$\ln y = x\ln(\cos x)$$

$$\frac{1}{y}\frac{dy}{dx} = x(-\tan x)+\ln(\cos x)$$

$$\frac{dy}{dx} = (\cos x)^x\left(\ln(\cos x)-x(\tan x)\right)$$

$$\Rightarrow g(x)=(\cos x)^x$$

$$g\left(\frac{\pi}{3}\right)=\left(\frac{1}{2}\right)^{\pi/3}\left(-\ln 2-\frac{\pi}{3}\sqrt{3}\right)$$

$$=-\left(\frac{1}{2}\right)^{\pi/3}\left(\ln 2+\frac{\pi}{\sqrt{3}}\right)$$

$\alpha = \ln 2+\frac{\pi}{\sqrt{3}} \Rightarrow e^{\alpha} = e^{\ln 2+\pi/\sqrt{3}}$

$= e^{\ln 2}\cdot e^{\pi/\sqrt{3}}$

$e^{\alpha} = 2\cdot e^{\pi/\sqrt{3}} \Rightarrow \frac{e^{\alpha}}{2} = e^{\frac{\pi}{\sqrt{3}}}$

10.

(A) Image of Point $(1,2,3)$

The formula for the image (x_i, y_i, z_i) of a point (x_1, y_1, z_1) with respect to a plane $ax+by+cz+d=0$ is:

$$\frac{x_i-x_1}{a}=\frac{y_i-y_1}{b}=\frac{z_i-z_1}{c}=-2\frac{ax_1+by_1+cz_1+d}{a^2+b^2+c^2}$$

Plugging in the values:

$$\frac{x_i-1}{1}=\frac{y_i-2}{1}=\frac{z_i-3}{1}=-2\frac{1(1)+1(2)+1(3)-3}{1^2+1^2+1^2}=-2\left(\frac{3}{3}\right)=-2$$

- $x_i-1=-2 \Rightarrow x_i=-1$
- $y_i-2=-2 \Rightarrow y_i=0$
- $z_i-3=-2 \Rightarrow z_i=1$

The image is $(-1,0,1)$. Statement (A) is True.

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11.

$z=3e^{i\theta};\omega=e^{i\theta};|\omega|=1 \quad \bar{\omega}^2=e^{-i2\theta}$

$\omega^2=e^{12\theta}$

$\bar{\omega}=e^{-i\theta} \quad \omega-\bar{\omega}=2i\sin\theta$

$z=3\omega$

$\left|g\omega^2-6i\omega+1\right|^2=\left(g\omega^2-6i\omega+1\right)\left(g\omega^{-2}+6i\bar{\omega}+1\right)$

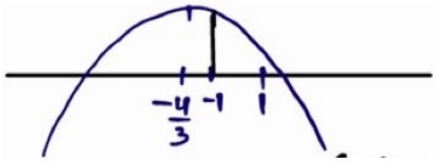
$\Rightarrow =118+48i\left(\omega-\bar{\omega}\right)+g\left(\omega^2+\bar{\omega}^2\right)$

$=118-96\sin\theta+18\cos 2\theta$

$f(\theta)=18\left(1-2\sin^2\theta\right)-96\sin\theta+118$

$=-36\sin^2\theta-96\sin\theta+136$

$\frac{-b}{2a}=\frac{+96}{-2.36}=\frac{-4}{3};\sin\theta=[-1,1]$



$f(\theta)_{max} \Rightarrow \sin\theta=-1 \qquad f(\theta)_{min}\ \sin\theta=1 \ =4$

$f(\theta)_{max}=-36+96+136=196$

$\left|9\omega^2-6i\omega+1\right|=\sqrt{196}=14$

$\left|9\omega^2-6i\omega+1\right|=\sqrt{4}=2$

12.

$$\left[\bar{p}\,\bar{q}\,\bar{r}\right]=\left[\frac{\bar{b}\times\bar{c}}{v}\ \frac{\bar{c}\times\bar{a}}{v}\ \frac{\bar{a}\times\bar{b}}{v}\right]$$

$$=\frac{1}{v^3}\left[\bar{b}\times\bar{c}\ \bar{c}\times\bar{a}\ \bar{a}\times\bar{b}\right]$$

$$=\frac{\left[\bar{a}\,\bar{b}\,\bar{c}\right]^2}{\left[\bar{a}\,\bar{b}\,\bar{c}\right]^3}=\frac{1}{\left[\bar{a}\,\bar{b}\,\bar{c}\right]}$$

$\rightarrow v$ is scalar

$$\bar{p}\times\bar{q}=\frac{\bar{b}\times\bar{c}}{v}\times\frac{\bar{c}\times\bar{a}}{v}$$

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$$=\frac{1}{v^2}\left(\left(\bar{b}\times\bar{c}\right)\times\left(\bar{c}\times\bar{a}\right)\right)$$

$$\bar{a}\times\left(\bar{b}\times\bar{c}\right)=\left(\bar{a}.\bar{c}\right)\bar{b}-\left(\bar{a}.\bar{b}\right)\bar{c}$$

Let $\bar{b}\times\bar{c}=\bar{u}$

Then $\left(\bar{u}\times\left(\bar{c}\times\bar{a}\right)\right)=\left(\bar{u}.\bar{a}\right)\bar{c}-\left(\bar{u}.\bar{c}\right)\bar{a}$

$$=\left[\bar{b}\,\bar{c}\,\bar{a}\right]\bar{c}-\left[\bar{b}\,\bar{c}\,\bar{c}\right]\bar{a}$$

$$=\left[\bar{a}\,\bar{b}\,\bar{c}\right]\bar{c}$$

$$\bar{p}\times\bar{q}=\frac{\left[\bar{a}\,\bar{b}\,\bar{c}\right]\bar{c}}{\left[\bar{a}\,\bar{b}\,\bar{c}\right]^2}=\frac{\bar{c}}{v}$$

$$\bar{q}\times\bar{r}=\frac{\bar{a}}{v}$$

$$\bar{r}\times\bar{p}=\frac{\bar{b}}{v}$$

13.

$$\frac{1}{r+1}\binom{n}{r}=\frac{1}{n+1}\binom{n+1}{r+1}$$

$$S=\sum_{r=0}^{n}\frac{1}{n+1}\binom{n+1}{r+1}\binom{n}{n-r}$$

$$(n+1)S=\sum_{r=0}^{n}\binom{n+1}{r+1}\binom{n}{n-r}$$

$$(r+1)+(n-r)=n+1.$$

The sum equates to $\binom{(n+1)+n}{n+1}=\binom{2n+1}{n+1}$.

$$11S={}^{21}C_{11}$$

$$\Rightarrow\frac{21!}{11!10!}\text{ power of }S=\frac{4}{2.2}\Rightarrow\text{Net power of }5=0\Rightarrow\text{no. zeros.}$$

14. $\left(x^2-1\right)2\sin x\cos x+2x\left(\cos^2 x-\sin^2 x\right)$

$$x^2\,2\sin x\cos x-2\sin x\cos x+2x\cos^2 x-2x\sin^2 x$$

$$2x\sin x\left(x\cos x-\sin x\right)+2\cos x\left(x\cos x-\sin x\right)$$

$$\left(x\cos x-\sin x\right)\left(2\cos x+2x\sin x\right)$$

$$=2\left(x\cos x-\sin x\right)\left(\cos x+x\sin x\right)$$

Let say $u=x\cos x-\sin x$

$$u'=x\left(-\sin x\right)+\cos x-\cos x$$

$$=-x\sin x$$

$$v=\cos x+x\sin x$$

$$v'=-\sin x+x\cos x+\sin x$$

$$v'=x\cos x\;\;uv'=\left(x\cos x-\sin x\right)x\cos x$$

$$=x^2\cos^2 x-x\sin x\cos x$$

$$u'v=-x\sin x\left(\cos x+x\sin x\right)$$

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$= -x\sin x\cos x - x^2\sin^2 x$

$uv' - vu' = x^2$

$\frac{1}{2}\int \frac{uv' - vu'}{uv}dx$

$\frac{1}{2}\int \frac{v'}{v} - \int \frac{u'}{u}$

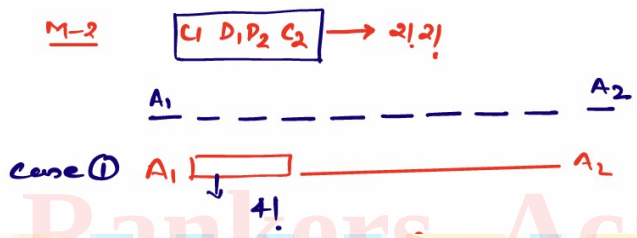
$\left(\frac{1}{2}\ln\left(\frac{\cos x + x\sin x}{x\cos x - \sin x}\right)\right)_{\frac{\pi}{2}}^{\pi}$

$= \frac{1}{2}\ln\left|\left(\frac{-1}{-\pi}\right)\right| - \frac{1}{2}\ln\left|\left(\frac{\frac{\pi}{2}}{-1}\right)\right| = \frac{1}{2}\left(\ln\left(\frac{1}{\pi}\right) - \ln\left(\frac{\pi}{2}\right)\right)$

$= \frac{1}{2}\ln\left(\frac{1}{\pi}\frac{2}{\pi}\right) = \ln\left(\frac{\sqrt{2}}{\pi}\right)$

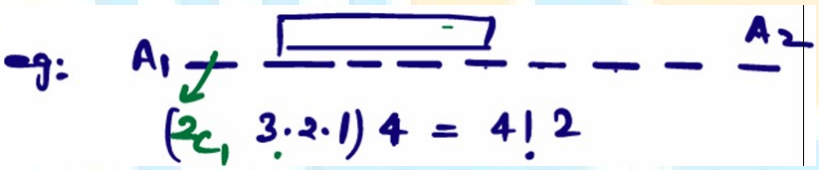
$k = \frac{\sqrt{2}}{\pi}$

15.



Box is adjacent A_1

Case (2) Box is not adjacent to A_1



$= (4!.3)\ 2!\ 2!\ 2! = 24.4!$

Total

$\Rightarrow A_1A_2\ D_1D_2\ C_1C_2$

Box $\boxed{R_1\ D_1\ D_2\ R_2}$ 2! 2!

$\times$

$\underline{A_1}\ \boxed{Box}$

(4 ways) for Box

$(4!)\ (4)\ 2!\ 2!\ 2!$

$A_1A_2\ D_1D_2\ R_1R_2$

$= 32(4!)$

Total $= (56)(4!) = 1344$

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16.

$$\sum_{k=0}^{50} 2^{50-k}\,{}^{50+k}C_k$$

$$2^{50}\sum_{k=0}^{50}\left(\frac{1}{2}\right)^k {}^{50+k}C_k = 2^{100}$$

$$\Rightarrow \log_8^{2^{100}} = \frac{100}{3} = 33.33$$

Note :-

$$\sum_{k=0}^{n} {}^{n+k}C_k\left(\frac{1}{2}\right)^k = 2^n$$

17.

$a=0$

$f(0)+2f(b)=f(f(b))$

Let $b\in z$ *say for* $x\in z$

$f(0)+2f(x)=f(f(x))$ _____ (1)

$a=1$

$f(2)+2f(b)=f(f(1+b))$

$f(2)+2f(b)=f(0)+2f(1+b)$

Let say $f(2)+2f(x)=f(0)+2f(1+x)$

$x\in z$

$f(2)-f(0)=2f(1+x)-2f(x)$

$\Rightarrow \dfrac{f(2)-f(0)}{2}=f(1+x)-f(x)$

R.H.S is difference of consecutive terms = L.H.S (is constant)

$\Rightarrow f(x)=mx+n;$ is an arithmetic progression

$f(2a)+2f(b)=f(f(a+b))$

$m(2a)+n+2mb+2n=f(m(a+b)+n)$

$=f(ma+mb+n)$

$=m^2a+m^2b+mn+n$

$2am+n+2mb+2n=m^2a+m^2b+mn+n$

$2m(a+b)+3n=(a+b)m^2+mn+n$

$2m=m^2$ *and* $mn=2n$ $\quad m=0\Rightarrow n=0$

$m=0,m=2$ $\quad n=0,m=2$ $\quad m=2\Rightarrow n\in z$

$f(x)=0$ *and* $f(x)=2x+n$

$-8=2.2+n$

$n=-12$

For More Join: @IITJEE_Advanced

$f(x)=2x-12$

$y=2x-12$

$(2x-12)^2=32x$

$4(x-6)^2=32x$

$x^2-12x+36=8x$

$x^2-20x+36=0$

$x^2-2x-18x+36=0$

$x(x-2)-18(x-2)=0$

$x=2 \quad , \quad x=18$

$S_1=(2,-8),\ S_2=(18,24)$

$$Area=\frac{1}{2}\begin{vmatrix}0 & 0 & 1\\ 2 & -8 & 1\\ 18 & 24 & 1\end{vmatrix}1$$

$$=\frac{1}{2}\left|(1)(48+144)\right|$$

$$Area=\frac{192}{2}=96.$$

18. $\int_0^1 x^{k-1}dx=\left[\frac{x^k}{k}\right]_0^1=\frac{1}{k}$

$$P=\sum_{r=1}^{50}(-1)^{r-1}\,{}^{50}C_r\sum_{k=1}^{r}\frac{1}{k}$$

$$P=\sum_{r=1}^{50}(-1)^{r-1}\,{}^{50}C_r\sum_{k=1}^{r}\int_0^1 x^{k-1}dx$$

$$P=\sum_{r=1}^{50}(-1)^{r-1}\,{}^{50}C_r\int_0^1\left(\sum_{k=1}^{r}x^{k-1}\right)dx$$

$$P=\sum_{r=1}^{50}(-1)^{r}\,{}^{50}C_r\int_0^1\frac{1-x^r}{1-x}dx$$

$$=\frac{1}{1-x}\sum_{r=1}^{50}(-1)^r\,{}^{50}C_r(1-x^r)$$

use:- $\sum_{k=1}^{r}x^{k-1}=1+x+x^2+\cdots+x^{r-1}$

$$=\frac{1(x^{r-1})}{(x-1)}=\frac{1-x^{r-1}}{1-x}$$

For More Join: @IITJEE_Advanced

$$\sum_{r=1}^{n}(-1)^{r-1}\binom{n}{r}(1-x^r)=\sum_{r=1}^{n}(-1)^{r-1}\binom{n}{r}-\sum_{r=1}^{n}(-1)^{r-1}\binom{n}{r}x^r$$
$$=[1-(1-1)^n]-[1-(1-x)^n]$$
$$=1-1+(1-x)^n$$
$$=(1-x)^n$$

Substitute the simplified sum back into the integral:

$$p=\int_0^1\frac{(1-x)^{50}}{1-x}dx$$
$$=\int_0^1(1-x)^{49}dx$$

Let $u=1-x$, then $du=-dx$. When $x=0, u=1$; when $x=1, u=0$:

$$p=\int_1^0 u^{49}(-du)$$
$$=\int_0^1 u^{49}du$$
$$=\left[\frac{u^{50}}{50}\right]_0^1=\frac{1}{50}$$

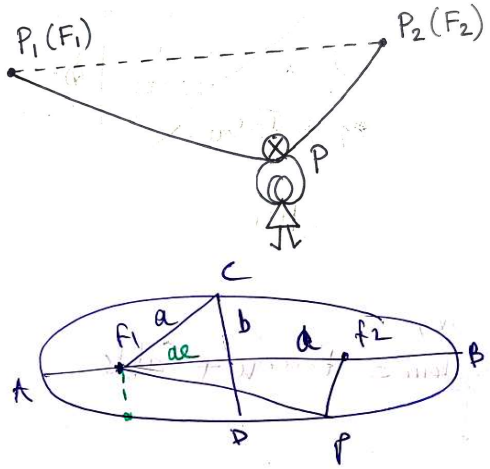
$$100p=100\times\frac{1}{50}=2$$

PHYSICS

19. Conceptual

20. Power $P\propto A(x).s(x)^2$ is constant. Since $A(x)\propto e^{-\lambda x}$, then $s(x)\propto e^{\lambda x/2}$. Solve for λx_1 using the given ratio.

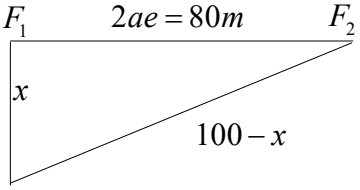
21. It is clear that the person will move along ellipse



$F_1P+F_2P=2a=100\text{ m}\Rightarrow\mathbf{a=50}\text{ m}$

$F_1F_2=2a=80\text{ m}\Rightarrow e=\frac{4}{5}\Rightarrow b=30\text{ m}$

Initial



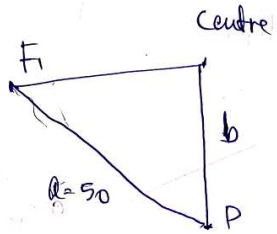
For More Join: @IITJEE_Advanced

$a(1-e^2)=18=$ Semi – latus rectum

$x^2+80^2=(100-x)^2$

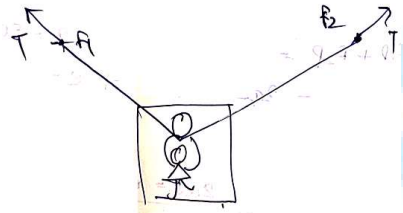
$x=18$ m

Bottom



Difference of Heights $30-18=12m$

Now consider, system = person + cable



F_1 and F_2 are fixed points.so points of application of T has no displacement

$w_T+w_{mg}=\frac{1}{2}mv^2-0$

$u^2=2g\Delta h=240\ \ m^2/s^2$

22. Let us consider Ampere's loop as shown:

$\oint \vec{B}\cdot d\vec{l}=0$

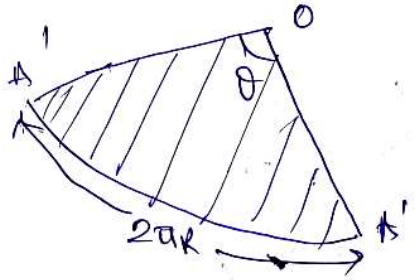
$\oint (\vec{B}_{st}+\vec{B}_{sp})\cdot d\vec{l}=0$

$\therefore B_{sp}=\frac{\mu_0 i}{4\pi R\sqrt{2}}(\sin 90^\circ-\sin\theta)$

$=\frac{\mu_0 i}{4\pi R}\left(\frac{1}{\sqrt{2}}-\frac{1}{\sqrt{6}}\right)$

23.

: when a cone is cut open, it forms sector

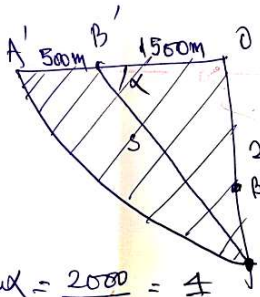


OA = Slant length = 2000km

R = 500m

$\theta=\frac{2\pi\times 500}{2000}=\pi/2$

For More Join: @IITJEE_Advanced



Let length of shortest path is S

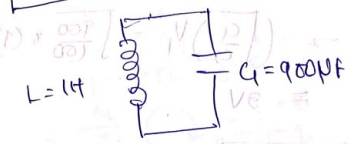
$\tan\alpha = \frac{2000}{1500} = \frac{4}{3}$

$\Rightarrow \alpha = 53^0$

$S = 2000 \operatorname{cosec} 53^0 = 2500m$ Ans

24.

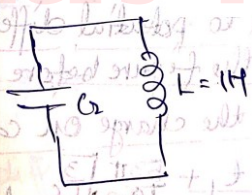
$0 < t < \frac{1.5\pi}{100}$ s



$T_1 = 2\pi\sqrt{LC} = \frac{6\pi}{100}$ s

At $t = \frac{1.5\pi}{100}$ s $= T_1/4$, the inductor gave all the energy which was stored in the capacitor C

$t > \frac{1.5\pi}{100}$ s



Now, $T_2 = 2\pi\sqrt{LC_2} = \frac{2\pi}{100}$ s. After $\frac{0.5\pi}{100}$ s which is $T_2/4$, the whole energy of the inductor gets transferred to the capacitor C_2

$\therefore \frac{1}{2}CV_1^2 = \frac{1}{2}C_2V_2^2$

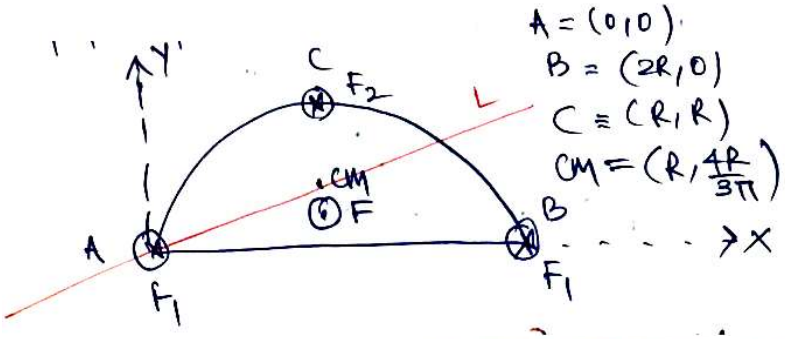
$\Rightarrow V_2 = \left(\sqrt{\frac{C}{C_2}}\right)V_1 = \sqrt{\frac{900}{100}} \times (1)$

$= 3V$

$\therefore \boxed{V_2 = 3V}$ Ans

For More Join: @IITJEE_Advanced

25. Let F_1 be force is red A and red B (by symmetry) and F_3 is red C. Let us consider FBP of front plate

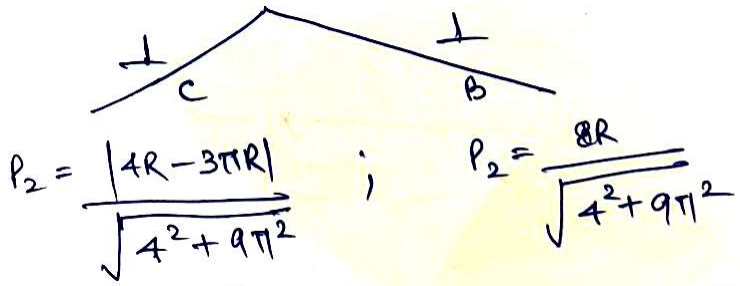


F = Force exerted by back plate on front plate

$$=\left(\frac{q_1}{2A\in_0}\right)q_2=\frac{q^2}{2A\in_0}=F$$

$\therefore 2F_1+F_2=F.....(1)$ equation of line joining (A) and (CM)

$$\boxed{y=\frac{4}{3\pi}x\Rightarrow 4x-3\pi y=0}$$



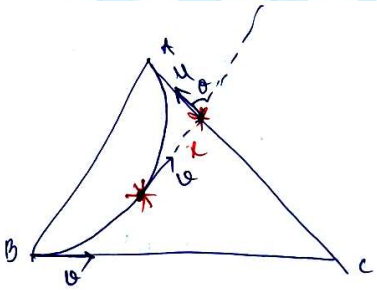
Now, torque/L = 0

$\Rightarrow F_2P_2=F_1P_1\Rightarrow$ strains are equal as plates remain parallel are equal (Marke's law)

$Y_A(8)=Y_B(8)=(3\pi-4)\lambda C$

Hence option (C) and (D) are correct

26.



Let x is separation between two ants

$$\therefore\frac{-dx}{dt}=v-u\cos\theta$$

$$\int_d^0-dx=\int_0^{T=\Delta t}(v-u\cos\theta)\,dt$$

$$\Rightarrow d=vT-u\int_0^T\cos\theta\,dt.....(1)$$

Or $\boxed{d=u\left(\frac{d}{u}\right)-u\int_0^T\cos\theta dt}$

For More Join: @IITJEE_Advanced

Along direction AC displacement of out from (B) is $\frac{d}{2}$

$$\boxed{\int_0^T V\cos\theta\, dt = \frac{d}{2}} \ldots\ldots\ldots(2)$$

From equation (1) and (2), we get

$$d = \frac{vd}{u} - u\left(\frac{d}{2v}\right)$$

$$1 = \frac{u}{u} - \frac{u}{2v}$$

Let $\frac{v}{u} = K$

$$1 = K - \frac{1}{2K} \Rightarrow 2K^2 - 2K - 1 = 0$$

$$\Rightarrow K = \frac{2+\sqrt{12}}{4} \Rightarrow v = \left(\frac{\sqrt{3}+1}{2}\right)u$$

$$\Rightarrow \boxed{vT = \left(\frac{\sqrt{3}+1}{2}\right)d}$$

For option (C)

$$\vec{F} = q(\vec{v}\times\vec{B})$$

$$\langle\vec{F}\rangle = q(\langle\vec{v}\rangle\times\vec{B})$$

$$= q\left(\frac{\vec{s}}{T}\times\vec{B}\right)$$

$$|\langle\vec{F}\rangle| = qvB_0 \quad \left(\frac{s}{T} = \frac{d}{T} = v\right)$$

For option (d) ant is moving in curvilinear path with constant speed

$\therefore a_t = 0,\ a_u \neq 0$

Hence option (A), (C) and (D) are correct

27. We know that for a rolling body

$$a_{cm} = \frac{g\sin\theta}{\alpha+\beta},\ f_{cm} = \beta mR^2$$

$\rho = \frac{1}{2}a_{cm}t^2 =$ length of track for greater β, less a_{cm}, more time taken $\beta_A = \frac{1}{2}$; $\beta_B = \frac{3}{10}$ for C,

moment of inertia $\beta_C(m_1+m_2)R^2 = \beta_A mR^2 + \beta_B m_2 R^2$

$$\Rightarrow \beta_C = \frac{\beta_A m_1 + \beta_B m_2}{m_1+m_2}(\text{weighted average})$$

$\Rightarrow \beta_C$ lies between $\Rightarrow \beta_A, \beta_B$

$\therefore \beta_B < \beta_C < \beta_A$

$\Rightarrow B < C < A$

For More Join: @IITJEE_Advanced

28.

$E_3 - E_2 = 47.25$ eV

$\Rightarrow -13.6\, Z^2 \left(\frac{1}{9} - \frac{1}{4}\right) = 47.25$

$\Rightarrow Z^2 = 25$

$\Rightarrow Z = 5$

(b) $E_4 - E_3 = -13.6 \times 25 \times \left(\frac{1}{16} - \frac{1}{9}\right)$

$= 16.53$ eV

(c) $n = 1 \rightarrow n = \infty$

$E = -13.6 \times 25\left(\frac{1}{\infty} - \frac{1}{1}\right) = 340$ eV

Wavelength required $= \frac{12400}{340}$ Å $= 36.56$Å

(d) $L = \frac{nh}{2\pi} = \frac{2 \times 6.62 \times 10^{-34}}{2 \times 3.14} = 2.19 \times 10^{-34}$

29.

$T_0 = \frac{mv_0^2}{r_0} \quad ---(1)$

$K_0 = \frac{1}{2}mv_0^2 \quad ---(2)$

COAM gives

$\vec{L_i} = \vec{L_f}$

$mv_0 r_0 = mvr \quad ---(3)$

From eq (1), (2) and (3) we get

$T = \frac{mv^2}{r} = \frac{m}{r}\left(\frac{v_0 r_0}{r}\right)^2$

$T = \frac{T_0 r_0^3}{r^3}$

$K = \frac{1}{2}mv^2 = \frac{1}{2}m\left(\frac{v_0 r_0}{r}\right)^2$

$K = \frac{K_0 r_0^2}{r^2}$

Also,

$T = T_0 \frac{r_0^3}{r^3}$

$\frac{T}{T_0} = \left(\frac{r_0}{r}\right)^3 \Rightarrow \frac{r_0}{r} = \left(\frac{T}{T_0}\right)^{1/3}$

$T = 40$ N, $T_0 = \frac{mv_0^2}{r_0} = \frac{(0.25)(4)^2}{0.8}$

$= 5$ N

$\therefore \frac{r_0}{r} = (8)^{1/3} = 2$

$\Rightarrow r = r_0 / 2 = 0.4$ m

Work done by tension = Change in Kinetic energy

For More Join: @IITJEE_Advanced

$= KE_f - KE_i$

$KE_i = K_0, \quad KE_f = K_0(2)^2 = 4K_0$

$\therefore$ Work done $= 3K_0$

30. Given $n_1 >> (n_1 - n_2) = \Delta n$

$\therefore P_1 = \frac{n_1 RT}{N_A}, P_2 = \frac{n_2 RT}{N_A}$

$\Rightarrow F = (n_1 - n_2) K_B TS = \Delta n K_B TS$

$V = \frac{\Delta n K_B TS}{\beta}$

$\Delta n K_\beta TS = \ln_1 S\beta V$

$\Rightarrow x_1 \beta V l = \Delta n K_\beta T$

Total number of molecules/second

$= \frac{(n_1 V dt)s}{dt} = n_1 VS = \frac{\Delta n K_\beta TVS}{\beta V l}$

$= \left(\frac{\Delta n}{l}\right)\left(\frac{K_\beta T}{\beta}\right) S$

As Δn null decrease with time, therefore rate of molecules getting transfer decrease with time

31. Normal component of velocities is same for surfaces maintaining contact.

Now, velocity component along the rod should be same

$2v\cos 45^\circ - v\cos 45^\circ = v\cos 45^\circ$

$\Rightarrow v_y = v$

$\vec{v}_A = 2v\hat{i} + v\hat{j}$

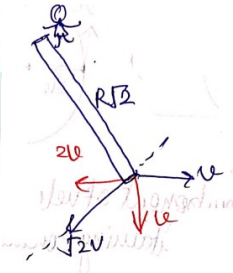
$\vec{v}_B = v\hat{i}$

$\vec{r}_{cm} = \frac{\vec{r}_A + \vec{r}_B}{2} \Rightarrow \vec{v}_{cm} = \frac{\vec{v}_A + \vec{v}_B}{2}$

$\vec{v}_M = \frac{3v}{2}\hat{i} + \frac{v}{2}\hat{j}$

$\omega_{BA} = \frac{\sqrt{2}v}{\sqrt{2}R} = \frac{v}{R}$

Note that relative velocity of two material points in GPM is perpendicular to line joining.



$KE = KE_{cm} + KE_{about\ cm}$

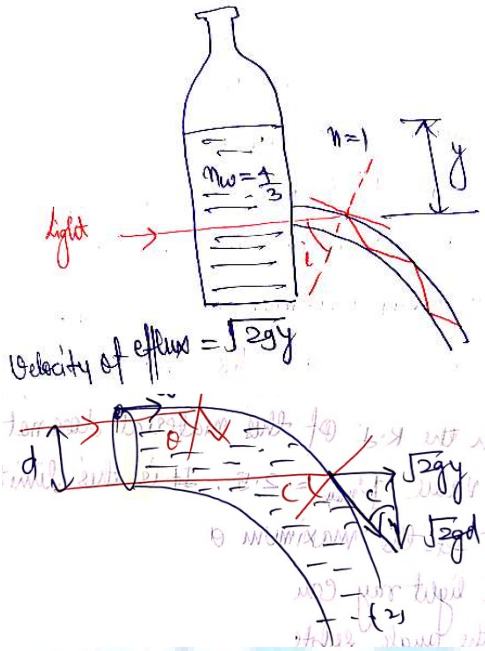
$= \frac{1}{2} m\left(\frac{9v^2}{4} + \frac{v^2}{4}\right) + \frac{1}{2}\frac{M(\sqrt{2}R)^2}{12}\left(\frac{v}{R}\right)^2$

$KE = \frac{4mv^2}{3}$

$\therefore K = 3$

For More Join: @IITJEE_Advanced

32.



$$\sin c = \frac{1}{\mu} = \frac{3}{4} \quad \ldots(1)$$

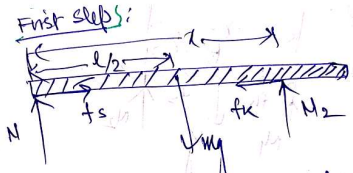
$$\tan c = \frac{\sqrt{2gy}}{\sqrt{2gd}} = \sqrt{\frac{y}{d}} \quad \ldots(2)$$

From eq (1) and (2)

$$\frac{\sqrt{y}}{\sqrt{y}+\sqrt{d}} = \frac{3}{4}$$

$$y = \frac{18}{7} \text{ mm}$$

33.



Torque balance about centre of rod gives

$$N_1 \ell/2 = N_2(x_f - \ell/2)$$

$$\mu N_1 = \frac{\mu}{3} N_2$$

$$N_2 x = mg\frac{\ell}{2}$$

$$\Rightarrow N_2 = mg\ell/2x$$

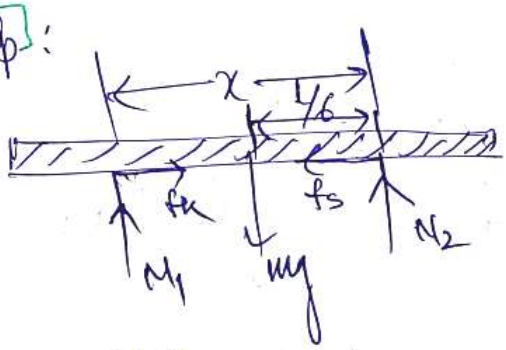
$$f_k = \mu_k N_2 = \frac{\mu mg\ell}{6} \propto \frac{1}{x}$$

$$\therefore W_f = \frac{\mu mg\ell}{6} \int_{\ell}^{2\ell/3} \frac{dx}{x}$$

$$\therefore W_f = -\frac{\mu mg\ell}{6}\left[\ln 3 - \ln 2\right]$$

For More Join: @IITJEE_Advanced

Second step:



$mg\frac{l}{6} = N_1 x$

$N_1 = \frac{mgl}{6x}$

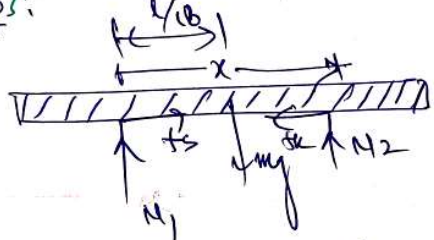
$x_f = \frac{l}{6} + \frac{1}{3}\left(\frac{l}{6}\right) = \frac{2l}{9}$

$x_i = \frac{2l}{3}$

$W_f = \int_{2l/9}^{} \left(\frac{\mu}{3}\right) N_1 dx$

$= \frac{\mu mgl}{18}\int_{2l/3} \frac{dx}{x} = -\frac{\mu mgl \ln 3}{18}$

Third step:



$mg \times \frac{l}{18} = N_2 x \Rightarrow N_2 = \frac{mgl}{18x}$

$x_f = \frac{l}{18} + \frac{1}{3}\left(\frac{l}{18}\right) = \frac{2l}{27}$

$x_i = \frac{l}{18} + \frac{l}{6} = \frac{2l}{9}$ $2l/27$

$W_f = \int \left(\frac{\mu}{3}\right) N_2 dx = \frac{\mu mgl}{18}\int_{2l/9} \frac{dx}{x}$

$= -\frac{\mu mgl}{54} \ln 3$

For More Join: @IITJEE_Advanced

Total work done by friction is given by

$$W_f = -\frac{\mu mgl}{6}\left[-\ln 2 + \ln 3\left(1+\frac{1}{3}+\frac{1}{3^2}+\ldots\right)\right]$$

$$\boxed{W_f = -\frac{\mu mgl}{6}\left[3\ln 3 - \ln 2\right]}$$

34.

$$C_{eq} = \frac{C_1C_2}{C_1+C_2} = 1200\,pF$$

$$\frac{\Delta C_{eq}}{C_{eq}^2} = \frac{\Delta C_1}{C_1^2} + \frac{\Delta C_2}{C_2^2}$$

$$\Rightarrow \Delta C_{eq} = 6\,pF$$

Now, $U = \frac{1}{2}CV^2$

$$100\times\frac{\Delta U}{U} = \left(\frac{\Delta C}{C} + 2\frac{\Delta V}{V}\right)\times 100 = 1.3\%$$

35.

$$h_1 + h_2 = 0.29\times 2 + 0.1 = 0.68 \quad \ldots (1)$$

$$\rho_k g(0.1) + \rho_w g h_1 - \rho_w g(0.1) = \rho_w g h_2$$

$$\Rightarrow h_1 - h_2 = 0.02 \quad \ldots (2)$$

From eq(1) and (2), we get

$$\frac{h_1}{h_2} = \frac{35}{33}$$

36. Conceptual

CHEMISTRY

37.

$$\lambda \propto \frac{1}{\sqrt{KE}}$$

$$= \sqrt{\frac{KE_1}{KE_2}} = \frac{\lambda_2}{\lambda_1}$$

$$= \frac{0.99\lambda_1}{\lambda_1}$$

$$KE_1 = (0.99)^2\, KE_2$$

$$KE_2 \approx 1.02\, KE_1$$

$$\%\text{ change is KE} = \frac{KE_2 - KE_1}{KE_1}\times 100$$

$$\approx 2\%$$

For More Join: @IITJEE_Advanced

41.

	ΔH (KJ/mole)
(i) $H(g) + e \rightarrow H^-(g)$	–73
(ii) $Li(g) + e \rightarrow Li^-(g)$	–60
(iii) $Na(g) + e \rightarrow Na^-(g)$	–53
(iv) $K(g) + e \rightarrow K^-(g)$	–48
(v) $P(g) + e \rightarrow P^-(g)$	–72
(vi) $O(g) + e \rightarrow O^-(g)$	–141
(vii) $O^-(g) + e \rightarrow O^{2-}(g)$	–798
(viii) $N(g) + e \rightarrow N^-(g)$	+ 6.8
(ix) $Ar(g) + e \rightarrow Ar^-(g)$	+ 34

42. (a), (b), (d), (e) & (h) are incorrect

44.

$$NH_4NO_3(aq) \longrightarrow N_2O + H_2O$$

$$NH_2OH + HNO_2 \longrightarrow N_2O + 2H_2O$$

Hence, (A) and (D) are the correct answers.

46.

$$2CaO \cdot 3B_2O_3 + 2Na_2CO_3 \longrightarrow 2CaCO_3\downarrow + Na_2B_4O_7 + 2NaBO_2$$

$$* \ Na_2B_4O_7 \xrightarrow{\Delta} \underbrace{NaBO_2 + B_2O_3}_{\text{Glassy lead}}$$

$$* \ CuSO_4 + B_2O_3 \xrightarrow[\text{flame}]{\text{Non-luminous}} Cu(BO_2)_2 + SO_3\uparrow$$

(Blue green)
(Cupric meta borate)

$$* \ Na_2B_4O_7 + 7H_2O \rightarrow \underbrace{2Na[B(OH)_4] + 2H_3BO_3}_{\text{buffer}}$$

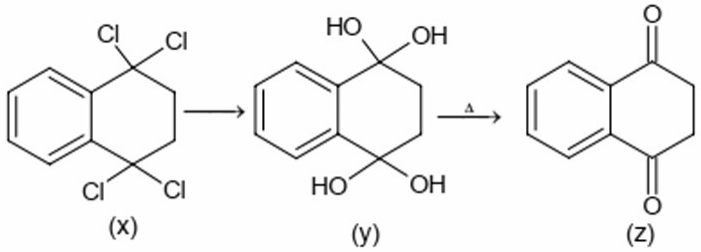
* In borax, 2B are sp^2 hybridised and 2B are sp^3 hybridised

For More Join: @IITJEE_Advanced

47. As Ni is added, CO will be consumed to attain the third equilibrium. As concentration of CO decreases, $COCl_2$ will decompose to a higher extent and hence Cl_2 concentration increases which in turn decreases the concentration of PCl_3.

50. $3OH^- + 3H_2O + P_4 \rightarrow PH_3 + 3H_2PO_2^-$

52.



(x) (y) (z)

53.

$A = 2y^{1/3}$ nm. $M = 6.023 \times y \times 4$ amu

$$\text{Density} = m/a^3 = \frac{6.023 \times 4 \times 1.66 \times 10^{-27}}{8y \times 10^{-27}} \text{kg/m}^3$$

$$= \frac{40}{8} = 5 \text{ kg/m}^3$$

54.

$$\Lambda_m = \frac{1000 \times 1.53 \times 10^{-4}}{0.0306} = 5 \text{ Scm}^2\text{mol}^{-1}$$

$B + H_2O \rightleftharpoons BH^+ + OH^-$

$\Lambda_m^\infty = 250$

$$\alpha = \frac{5}{250} = \frac{1}{50}$$

$$K_b = C\alpha^2 = 0.0306 \times \left(\frac{1}{50}\right)^2 = 1.224 \times 10^{-5}$$

$$= 12.24 \times 10^{-6}$$

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